Cartesian product:

The Cartesian product of two sets A and B denoted by $A \times B$ in the set of all ordered pairs of the form (a, b) where $a \in A$, $b \in B$.

$$A \times B = \{(a, b) \mid a \in A, b \in B\}$$

Ex:

① If $A = \{1, 2\}$, $B = \{x\}$

then $A \times B = \{(1, x), (2, x)\}$

② If $A = \{a, b, c\}$, $B = \{1, 2\}$

then $A \times B = \{(a, 1), (a, 2), (b, 1), (b, 2), (c, 1), (c, 2)\}$

$B \times A = \{ (1,a), (1,b), (1,c), (2,a), (2,b), (2,c) \}$

Note: $A \times B \neq B \times A$.

Binary relation:

When A and B are sets, a subset R of the Cartesian product $A \times B$ is called a binary relation from A to B.

If R is a binary relation from A to B and if $(a,b) \in R$, where $a \in A$ and $b \in B$ then we say that 'a' is related to 'b' and is written as aRb.

If $(a,b) \notin R$ it is denoted by $a \not R b$.

Note:
Mostly we will deal with relationships between the elements of two sets, Hence the word 'binary' will be omitted hereafter.

If R is a subset of $A \times A$ then R is called a relation on the set A.

The set $\{a \in A \mid aRb, \text{ for some } b \in B\}$ is called the domain of R and denoted by D(R).

The set $\{b \in B \mid aRb, \text{ for some } a \in A\}$ is called the range of R denoted by R(R).

Ex:

(1) Let $A = \{0, 1, 2, 3, 4\}$

$B = \{0, 1, 2, 3\}$ and aRb iff $a + b = 4$.

Then $R = \{(1,3), (2,2), (3,1), (4,0)\}$

The domain of $R = \{1, 2, 3, 4\}$
& the range of $R = \{0, 1, 2, 3\}$

(2) Let R be the relation on
$A = \{1, 2, 3, 4\}$ defined by
aRb if $a \leq b$, $a, b \in A$.

Then $R = \{(1,1), (1,2), (1,3), (1,4),$
$(2,2), (2,3), (2,4), (3,3), (3,4),$
$(4,4)\}$

The domain & range of R
are both equal to A.

Properties of Relations:

Let A be a set and R be a relation on A.

Then R is called

(i) Reflexive if for all $a \in A$, aRa

(ii) Symmetric if aRb holds $\Rightarrow$ bRa must be hold, for $a, b \in A$

(iii) Anti symmetric, if aRb and bRa hold then $a=b$.

(iv) Transitive, if aRb and bRc hold $\Rightarrow$ aRc must be hold, for $a, b, c \in A$.

(v) R is not reflexive, if $(a,a) \notin R$ for atleast $a \in A$
(irreflexive)

(vi) R is not symmetric if $(a,b) \in R$ but $(b,a) \notin R$ for at least one pair $(a,b) \in R$.

(vii) R is not transitive if $(a,b) \in R$, $(b,c) \in R$ but $(a,c) \notin R$ if $a, b, c \in A$.

(viii) R is not anti symmetric, if aRb and bRa but $a \neq b$.

Note:

If a relation which is not symmetric, it is not necessarily anti symmetric.

Ex:

If R is a relation on $A = \{1, 2, 3\}$ defined by $(a, b) \in R$ if $a \leq b$ where $a, b \in A$, then

$R = \{(1,1), (1,2), (1,3), (2,2), (2,3), (3,3)\}$

R is reflexive, since each of the elements of A is related to itself, as (1, 1), (2, 2) and (3, 3) are members in R.

If R is a relation on $A = \{1, 2, 3\}$ defined by $(a, b) \in R$ if $a < b$, where $a, b \in A$, then

$$R = \{(1, 2), (2, 3), (1, 3)\}$$

Here R is not reflexive (or) R is irreflexive.

The relation of perpendicularity on a set of lines in the plane is symmetric, since if a line a is perpendicular to the line b, then b is perpendicular to a.

The relation $\leq$ on the set Z of integers is not symmetric, since, for example, $4 \leq 5$, $5 \nleq 4$.

The relation of divisibility on N is antisymmetric, since whenever m is divisible by n and n is divisible by m then $m = n$.

Note:

Symmetry and anti symmetry are not negative of each other.
For example, the relation $R = \{(1,3), (3,1), (2,3)\}$ is neither symmetric nor anti symmetric, whereas, the relation $S = \{(1,1), (2,2)\}$ is both symmetric and anti symmetric.

The relation of set inclusion on a collection of sets is transitive since if $A \subseteq B$ and $B \subseteq C$, $A \subseteq C$.

Equivalence relation:-

A relation R on a set A is called an equivalence relation (or) RST relation if R is reflexive, symmetric and transitive.

Ex:

Let $A = \{a, b, c\}$ and a relation

$R = \{(a,a), (a,b), (b,b), (b,a), (a,c), (c,a), (c,c)\}$

R is reflexive, since $(a,a), (b,b), (c,c) \in R$ for all A.

R is symmetric, since $(a,b) \in R$, $(b,a) \in R$, $(a,c) \in R$, $(c,a) \in R$

Again R is transitive.

Hence R is an equivalence relation.

partial ordering or partial order relation:

A relation R on a set A is called a partial ordering or partial order relation if R is reflexive, anti symmetric and transitive.

ie R is a partial order relation on A if it has the following three properties.

1. aRa for every $a \in A$
2. aRb and $bRa \Rightarrow a=b$
3. aRb and $bRc \Rightarrow aRc$.

Partially ordered set or poset:

A set A together with a partial order relation R is called a partially ordered set or poset.

<u>Ex:</u> The greater than or equal to ($\geq$) relation is a partial odering on the set of integers Z, since 1. $a \geq a$ for every integer a,

i.e $\geq$ is reflexive

2. $a \geq b$ and $b \geq a \Rightarrow a = b$

i.e $\geq$ is anti symmetric

3. $a \geq b$ and $b \geq c \Rightarrow a \geq c$

i.e $\geq$ is transitive. Thus $(Z, \geq)$ a poset.

Partition of a set:

If S is a non-empty set, a collection of disjoint non empty subsets of S whose union is S is called a partition of S.

In other words, the collection of subsets A_i is a partition of S if and only if

(i) $A_i \neq \phi$, for each i

(ii) $A_i \cap A_j = \phi$, for $i \neq j$ and

(iii) $\bigcup_i A_i = S$, where $\bigcup_i A_i$ represents the union of the subsets A_i for all i.

Note:

The subsets in a partition are also called blocks of the partition.

Ex:

If $S = \{1, 2, 3, 4, 5, 6\}$

then

(i) $[\{1, 3, 5\}, \{2, 4\}]$ is not a partition, since the union of the subsets is not S, as the element 6 is missing.

(ii) $[\{1, 3\}, \{3, 5\}, \{2, 4, 6\}]$ is not a partition, since $\{1, 3\}$ and $\{3, 5\}$ are not disjoint.

(iii) $[\{1, 2, 3\}, \{4, 5\}, \{6\}]$ is a partition.